

RAMAKRISHNA KARNATI

Azure Cloud Data Engineer | ETL, DevOps, Visualization

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Summary

Results-driven **Azure Cloud Data Engineer** with **4+ years** of experience across **Cloud Infrastructure, ETL Design, and DevOps Automation**. Specialized in building scalable Azure data platforms using **Azure Data Factory (ADF), Databricks, ADLS Gen 2, and Fabric**. Proven expertise in developing **CI/CD pipelines**, optimizing cloud costs, and delivering high-performance data solutions processing **billions of records** efficiently on daily basis. Strong track record of improving system efficiency, enabling automation, and driving business insights through **data visualization**.

Key highlights include:

- Designed and developed scalable and metadata-driven ETL pipelines in ADF migrating **terabytes** of data from on-premise data sources to Azure, achieving a **99% success rate**.
- Developed **PySpark**-based data processing solutions handling **gigabytes** of data daily across multiple formats from various sources like **SAP, Salesforce, SFTP servers**, and **on-premise databases**.
- Led a team of 5** to implement an **LLM**-based solution to categorize **billions** of products into business provided classifications based on their descriptions and UPC codes, achieving over **90% accuracy**.
- Automated **Vacuum** and **Optimize** operations on Databricks tables, enhancing storage efficiency and boosting table performance by **40%**, and optimized query performance through improved code design and utilizing advanced techniques like **partitioning, broadcasting** and **liquid clustering**.
- Reduced over **85%** manual infrastructure setup time using **ARM templates, Terraform**, and **Azure DevOps** pipelines.
- Introduced a **selective deployment** framework within the **CI/CD** pipeline, enabling targeted artifact deployment for increased flexibility and reduced deployment time and integrated with **SonarQube** and **ARMTTK** tools to automate code quality checks, reducing manual review effort by **80%**.
- Built near real-time **Power BI** dashboards for ADF/Databricks pipeline monitoring, improving performance insights.

Skills

Programming: Python, PySpark, SQL, Java, Scala, PowerShell, Unix

Cloud Tools: Azure Data Factory, Databricks, ADLS, Logic Apps, Azure DevOps, Automation Account, Purview, Terraform, ARM Templates, Microsoft Fabric (Warehouse, Lakehouse, Data Engineering, Data Science), Power BI, AWS (EC2, S3)

Other Tools: Git, HTML, CSS, JavaScript, Spring Boot, Docker, Kubernetes, SonarQube

Soft Skills: Quick learning ability, Problem solving, Critical thinking, Attention to detail, Adaptability, Business communication

Experience

Specialist Programmer, Infosys, Hyderabad

Aug 2022 – Present

Cloud Infrastructure & DevOps:

- Provisioned full-stack Azure environments (ADLS, Databricks, ADF, VM) with private networking via **ARM Templates** and **Terraform**, cutting setup time to just around **20 minutes**, an **85%** reduction compared to manual configuration time.
- Designed and implemented end-to-end **CI/CD pipelines** in Azure DevOps using **YAML** and **Declarative Automation Bundles (DAB)**, enabling automated deployment of ADF pipelines, Databricks notebooks/workflows, and infrastructure components, aligned with standard **Git-based branching strategies** and release governance.
- Introduced a **selective deployment framework** within CI/CD pipeline, allowing targeted deployment of specific artifacts rather than the entire codebase, enhancing flexibility and reducing deployment time.
- Integrated **SonarQube** and **ARMTTK** code scanning tools into Azure DevOps pipelines to enforce infrastructure and code quality standards during deployment, reducing manual review effort by **80%**.
- Automated provisioning and configuration of Databricks clusters, pools, etc., along with **Microsoft Purview** data sources and scans using **REST APIs**, enhancing deployment efficiency by **70%** and streamlining data governance processes.
- Developed **PowerShell-based** automation leveraging Azure Automation Account and AZ modules to generate cost analytics and ADF/Databricks job performance reports, reducing manual effort by **95%** and enabling proactive cost optimization.

ETL & Data Engineering:

- Designed and built scalable and metadata-driven **ETL pipelines** using Azure Data Factory and Databricks, orchestrating ingestion of **terabytes of data** from SAP, Salesforce, SFTP, and RDBMS sources into Delta Lake with a **99% success rate**.
- Developed **PySpark-based** data processing frameworks to transform and load **GB–TB scale data** across multiple formats (CSV, JSON, Parquet, XML), enabling efficient batch processing for downstream analytics.
- Implemented **event-driven data ingestion architecture** using event based triggers in ADF, automating processing of **20+ daily file arrivals** and reducing manual intervention by **70%**.
- Engineered end-to-end ingestion workflows integrating **Azure Logic Apps, ADF, and Databricks** to extract SAP data via RFCs and process it into curated Delta Lake layers.
- Optimized **Delta Lake performance** by implementing **partitioning, liquid clustering, OPTIMIZE, and VACUUM**, improving

query performance by **40%** and reducing storage overhead.

- Built robust **logging and alerting frameworks** in ADF with custom error handling and **HTML**-based notifications, enabling faster incident resolution and pipeline observability.
- Developed high-throughput **JDBC-based** ingestion pipelines to load **millions of records daily** from on-prem databases into Delta Lake tables through Databricks workflows.
- Led a team of 5 to Design a large-scale data classification pipeline for **2.5B+ records** using hybrid approaches (rule-based, keyword, and **LLM-based via Azure OpenAI**), achieving over **90% accuracy**.
- Engineered a **cost-optimized LLM processing strategy** by sampling **0.2% of over 2 billion records** of data to generalize classification, significantly reducing compute costs while maintaining accuracy.
- Built unified & re-usable **Databricks workflows** to process both historical and incremental data, ensuring scalable and consistent data pipelines across ingestion layers.

Reporting & Visualization:

- Designed and developed centralized **Power BI** monitoring dashboard by ingesting **ADF pipeline logs** via Databricks tables, enabling near real-time monitoring of **pipeline health, failure rates, and SLA compliance** across **150+** pipelines.
- Designed and developed **AI/BI dashboards in Databricks** to monitor **LLM usage metrics**, including input/output token consumption, request volume, and cost analysis, enabling real-time visibility into model utilization.
- Built a **data reconciliation** framework in Power BI integrating data from **Azure SQL, Databricks, and SharePoint**, to detect discrepancies across systems and trigger notifications to respective stakeholders through **Power Automate**.
- Modeled optimized **star schema semantic layers** by de-normalizing complex relational datasets into fact and dimension tables, leveraging **DAX measures, relationships, and aggregations** to improve report query performance by **30%**.
- Automated **Power BI dataset refresh orchestration** using **REST APIs** integrated with ADF pipelines, enabling near real-time data availability post ETL completion and reducing data latency from **2 hours to 10 minutes**.

Data Governance:

- Implemented centralized **data governance using Unity Catalog**, managing **external and managed Delta tables** with a unified metastore, enabling fine-grained access control, data lineage tracking, and secure data sharing across environments.
- Established **data quality monitoring frameworks** for curated Delta tables, profiling datasets to track schema consistency, data freshness, and usage patterns over time.
- Designed and enforced **Row-Level Security (RLS)** on **100M+ record datasets** using **Azure AD group-based access control**, ensuring secure and role-based data access for downstream consumers.
- Integrated **data validation rules** within ETL pipelines to enforce business constraints, detect anomalies, and prevent propagation of invalid data into curated layers.

Senior Analyst Intern, Capgemini, Hyderabad

Feb 2022 – Apr 2022

- Worked with a team of 5 members to develop a **Java-based Spring application** for student attendance management using **GitHub** for version control and seamless collaboration.
- Utilized **PostgreSQL** as the DBMS and integrated the **JDBC API** into the application to perform efficient read/write operations on the database.
- Developed **JUnit** test cases to ensure robust functionality, achieving over 90% test coverage.
- Containerized the entire application using **Docker** and deployed it on an **AWS EC2 instance**, leveraging **Kubernetes**.

Awards & Achievements

- Recognized by internal delivery heads for representing my current project as the primary presenter during an external audit, delivering end-to-end walkthrough of architecture design, delivery processes, and documentation as per the comprehensive audit checklist, contributing to Infosys securing the [Data Warehouse Migration to Microsoft Azure specialization](#).
- Earned an official appreciation from the client for resolving a critical incident during a crucial month-end business closure period outside the scope of the project and regular work hours.
- Received a designated client appreciation for delivering exceptional support on a separate project, particularly for seamlessly automating CI/CD deployments.
- Consistently recognized with "**Rise Insta**" awards for exceptional performance every quarter and awarded with the "**Business Ninja Award**" in recognition of remarkable contributions and outstanding achievements.
- Achieved the "**Outstanding**" rating twice in year-end performance appraisals, reflecting exceptional contributions.

Certifications

- [Databricks Certified: Data Engineer Professional](#)
- [Microsoft Certified: Azure Solutions Architect Expert \(AZ-305\)](#)
- [Microsoft Certified: Fabric Data Engineer Associate \(DP-700\)](#)
- [Microsoft Certified: Fabric Analytics Engineer Associate \(DP-600\)](#)
- [Microsoft Certified: Azure Fundamentals \(AZ-900\)](#)
- [Microsoft Certified: Azure Data Fundamentals \(DP-900\)](#)

Education

B.Tech, Electronics & Communication Engineering

Vardhaman College of Engineering, Hyderabad

2018 – 2022